

Yiyuan Li

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github.com/yuann3

Summary

Backend Software Engineer with 2+ years of experience in systems programming, distributed systems, and full-stack development. Proficient in Go, Rust, Python, and C with hands-on experience building database engines, HTTP servers, and AI-powered applications. Built a Redis server implementation passing 100% of integration tests and a SQLite engine with B-tree indexing from scratch. Strong foundation in data structures, algorithms, and software architecture. Seeking backend or systems engineering roles to build scalable, high-performance solutions.

Technical Skills

Languages: Go (Golang), Rust, Python, C, C++, Java, SQL, JavaScript, TypeScript, Bash

Frameworks & Libraries: React.js, Next.js, FastAPI, Flask, Node.js, REST API, JWT

Databases & Tools: Redis, SQLite, PostgreSQL, MSSQL, ChromaDB, Docker, Git, Linux, CI/CD

Concepts: Distributed Systems, Microservices, RAG (Retrieval-Augmented Generation), API Design, Multithreading, Concurrency, Test-Driven Development (TDD), Agile

Experience

Software Engineering Intern, The University of Newcastle – Singapore Jan 2025 – Present

- Designed and developed a full-stack RAG (Retrieval-Augmented Generation) AI learning platform using FastAPI, React 19, and Ollama, enabling intelligent document Q&A for 500+ users
- Implemented document parsing pipeline with LlamaParse and vector search using ChromaDB, reducing query response time by 40%
- Built OAuth 2.0 authentication system and RESTful API endpoints handling 1,000+ daily requests
- Tech Stack: Python, TypeScript, React 19, Next.js, FastAPI, Flask, Tailwind CSS, JWT, ChromaDB

Peer Assisted Study Sessions (PASS) Leader, The University of Newcastle – Singapore Feb 2024 – Nov 2024

- Mentored 10+ students in Data Structures and Algorithms (DSA) and Object-Oriented Programming (OOP) using C++ and Java, improving average exam scores by 15%
- Delivered 10+ technical workshops on algorithm design, recursion, and sorting algorithms with live coding demonstrations

Projects

Sequel: SQLite Database Engine github.com/yuann3/sequel

- Built a SQLite-compatible database engine from scratch in Rust, implementing B-tree data structures and binary file parsing to read .db files without external dependencies
- Implemented SQL query execution including SELECT with WHERE clauses, COUNT aggregation, and index optimization achieving 3x faster lookups through B-tree traversal
- Technologies: Rust, B-tree, Binary Parsing, Database Internals, File I/O

Rego: Redis Server Implementation github.com/yuann3/Rego

- Implemented Redis-compatible server in Go with core commands (GET, SET, EXPIRE) and WAIT replication, passing 100% of Codecrafters integration tests
- Optimized memory allocation using slice pooling, reducing garbage collection overhead and improving throughput by 30%
- Technologies: Go (Golang), TCP/IP Networking, Concurrency, Memory Optimization

Ruskey: Programming Language Interpreter github.com/yuann3/Ruskey

- Developed a complete interpreter for Monkey language in Rust: lexer, recursive descent parser, AST, and tree-walking evaluator
- Implemented closures, first-class functions, and hash maps using Test-Driven Development (TDD) methodology
- Technologies: Rust, Compiler Design, Abstract Syntax Trees (AST), TDD

Pew: CLI Codebase Exporter

github.com/yuann3/Pew

- Built CLI tool in Go to export codebases to Markdown for LLM context and documentation workflows
- Implemented gitignore pattern matching and syntax-highlighted output with directory tree visualization
- Technologies: Go (Golang), CLI Development, File System Operations

Education

Ecole 42 Singapore, Singapore University of Technology and Design (SUTD)

Nov 2025 – Nov 2026

- Core Curriculum: Peer-to-peer software engineering program focused on C, Unix, and systems programming
- Completed 16+ projects: libft, ft_printf, get_next_line, push_swap, pipex, Born2beRoot

The University of Newcastle, Bachelor of Information Technology

Jan 2024 – Sep 2025

- High Distinction: Object-Oriented Programming (OOP)
- Distinction: Data Structures and Algorithms (DSA), Advanced Database Systems